TAREQ ABOUSHI

OFFICIAL PORTFOLIO
King Khalid Univ. • AI Track CS

Software Engineer • Database Architect • Systems Analyst

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Executive Profile & Alignment with Official CV: Senior Software Engineer, Database Architect, and Systems Analyst (B.Sc. in Computer Science - AI Track, King Khalid University). Professional background combining IT infrastructure and data management leadership with specialized cooperative training at the Saudi Electricity Company. Elite proficiency across a polyglot stack (Python, Go, TypeScript, Java, C#, SQL) delivering production enterprise systems, low-level storage engines (LSM-Tree/WAL), and statutory compliance suites (ZATCA Phase-2 & IFRS-16/18).

01	**Aseer Peaks Platform**	Software Engineering & GIS	HTML5, Tailwind, Leaflet GIS, WhatsApp API	● Live on Pages
02	**SovereignCFO Suite**	Systems Analysis & ERP Architecture	IFRS Double-Entry, ZATCA Phase-2, TLV Base64	● Live on Pages
03	**PulseKV Storage Engine**	Database Architecture & Internals	LSM-Tree, WAL, MemTable, Bloom Filter	● Live on Pages
04	**Axiom QuantFinance**	Quantitative Systems Engineering	Saudi Tadawul Engine, Monte Carlo, Greeks	● Live on Pages
05	**Qattat Code Training**	Compilers, AST & UI/UX Architecture	Interactive Sandbox, AST Runner, Asiri Heritage	● Live on Pages

PROJECT 01: ASEER PEAKS TOURISM & GIS PLATFORM | عسير السياحية والبيئية منصة قمم

[Software Engineering & Enterprise Web Platform]

System Scope: A premier cloud-native tourism and local artisan commerce ecosystem built specifically for the Asir region. Provides interactive geospatial exploration, authentic merchant listings, multi-category directory (souks, farms, dining), and automated WhatsApp commerce channels.

Architecture & Engineering Highlights:
- Full bilingual bidirectional engine (100% Arabic RTL & English LTR).
- Adaptive dual-mode GIS: Direct deep-linking to native Google Maps on mobile devices and an in-app interactive modal map with exact coordinate projection on desktop.
- Zero-latency client-side caching & instant category filtering.
- Direct-to-merchant transaction pipeline via automated WhatsApp URL scheme.

Key Performance Indicators & Specs:
- Lighthouse Score: 98/100 across Mobile & Desktop
- Rendering FPS: 60 FPS hardware-accelerated transitions
- Deployment: Automated GitHub Pages CI/CD
- Localization: 100% dynamic bilingual schema

Live Web Application: https://tareq0001.github.io/Aseer-Tourism-Platform/

GitHub Repo: Tareq0001/Aseer-Tourism-Platform

PROJECT 02: SOVEREIGNCFO ENTERPRISE ERP & ZATCA PHASE-2 | منظومة الإدارة المالية والفوترة المؤسسية

[Professional Systems Analysis & Enterprise Architecture]

System Scope: An institutional-grade financial enterprise resource planning (ERP) system architected for multi-entity corporate structures. Implements real-time IFRS-16 lease accounting, double-entry general ledger, and complete cryptographic compliance with Saudi ZATCA Phase-2 (FATOORA).

Architecture & Engineering Highlights:
- Compliant with Saudi ZATCA Phase-2 e-invoicing cryptographic specifications (ECDSA secp256k1 & TLV Base64 encoding).
- Automated standard QR code generator with cryptographic hash digests.
- IFRS-16 & IFRS-18 multi-entity double-entry balance sheet consolidation.
- Real-time FP&A; analytics with dynamic treasury liquidity forecasts.

Key Performance Indicators & Specs:
- ZATCA Compliance: Phase-2 Fatoora Cryptographic Stamp
- Accounting Engine: Strict Double-Entry Balanced Books
- Supported Entities: Unlimited Multi-Branch Consolidation
- Auditability: Immutable Transaction Ledger (Append-Only)

Live Web Application: https://tareq0001.github.io/SovereignCFO-Enterprise-Finance-Suite/

GitHub Repo: Tareq0001/SovereignCFO-Enterprise-Finance-Suite

PROJECT 03: PULSEKV STORAGE ENGINE & DATABASE INTERNALS | محرك قواعد البيانات المتطورة ومخزن المفتاح-القيمة

[Database Architect & Low-Level Storage Engines]

System Scope: A high-performance, fault-tolerant Key-Value database storage engine inspired by RocksDB and LevelDB. Designed from foundational principles to maximize sequential write throughput, minimize write amplification, and guarantee ACID durability through write-ahead logging.

Architecture & Engineering Highlights:
- Log-Structured Merge (LSM) Tree architecture optimizing random writes into sequential disk I/O.
- Crash-resilient Write-Ahead Log (WAL) ensuring zero data loss on power-cut scenarios.
- In-memory SkipList MemTable with concurrent multi-thread lock-free reads.
- Probabilistic Bloom Filter indexing eliminating over 99% of unnecessary SSTable disk seeks.
- Background leveled compaction worker reclaiming tombstoned keys.

Key Performance Indicators & Specs:
- Write Throughput: Exceeds 120,000 ops/sec sequential
- Bloom Filter False +: < 1% configured false positive rate
- Durability Level: Full ACID Durability (Strict WAL Flush)
- Compaction Strategy: Multi-tier Leveled Compaction

Live Web Application: https://tareq0001.github.io/PulseKV-Storage-Engine/

GitHub Repo: Tareq0001/PulseKV-Storage-Engine

PROJECT 04: AXIOM QUANTFINANCE & SAUDI TADAWUL ENGINE | المشتقات محطة التحليل المالي الكمي وهندسة

[Quantitative Systems Analysis & High-Performance Streaming]

System Scope: An institutional quantitative finance computing station and options pricing laboratory calibrated for the Saudi Stock Exchange (Tadawul). Performs ultra-fast Black-Scholes Greeks calculations, Monte Carlo risk path simulations, and Value-at-Risk (VaR) distribution modelling.

Architecture & Engineering Highlights:
- Real-time market data feed simulation mirroring Saudi Tadawul TASI ticker feeds.
- Closed-form Black-Scholes options valuation engine calculating Delta, Gamma, Theta, Vega, and Rho.
- Parallel Monte Carlo engine executing 10,000+ stochastic price trajectories in milliseconds.
- Historical Value-at-Risk (VaR 95% / 99%) and Maximum Drawdown risk matrix.

Key Performance Indicators & Specs:
- Simulation Paths: 10,000+ concurrent Monte Carlo runs
- Market Calibration: Saudi Equities (Tadawul TASI parameters)
- Latency: < 2ms calculation time for complex derivatives
- Visual Analytics: Interactive Canvas Surface Visualizer

Live Web Application: https://tareq0001.github.io/Axiom-QuantFinance/

GitHub Repo: Tareq0001/Axiom-QuantFinance

PROJECT 05: QATTAT CODE INTERACTIVE CODING ENGINE | التفاعلية والتراثية منصة قطاط البرمجة

[Software Engineering, Compilers & Educational Tech]

System Scope: An innovative interactive coding academy combining modern software engineering and abstract syntax tree (AST) execution with Saudi national culture (Asiri Qatt Art). Evaluates algorithmic problem-solving in real-time within an isolated in-browser sandbox.

Architecture & Engineering Highlights:
- Sandboxed browser execution environment with real-time console I/O streaming.
- AST algorithmic validator checking code complexity, syntax compliance, and logic flow.
- Gamified multi-tier progression mapping algorithmic mastery to authentic Asiri cultural patterns.
- Ultra-clean responsive IDE interface with multi-language code highlighting.

Key Performance Indicators & Specs:
- Execution Model: Client-Side Sandboxed Evaluation
- Supported Languages: Python, JavaScript, Computational Logic
- Design Integration: Authentic UNESCO-Recognized Qatt Art
- Target Audience: Saudi Vision 2030 Digital Skills Developers

Live Web Application: https://tareq0001.github.io/Qattat-Code/

GitHub Repo: Tareq0001/Qattat-Code

BONUS FLAGSHIP SHOWCASE: CUPERTINO-TITANIUM-PRO (APPLE HIG INDUSTRIAL DESIGN)

[Frontend & Spatial UI]

A flagship demonstration of Apple Human Interface Guidelines (HIG) engineering. Showcases custom glassmorphism, aerospace-grade titanium specular reflections, and GPU-accelerated fluid physics.
Live URL: https://tareq0001.github.io/Cupertino-Titanium-Pro/ | Repo: Tareq0001/Cupertino-Titanium-Pro